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GENERATIVE AI PROJECT



PocketSmart AI

Your Smart Budget & Recommendation Assistant

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Course	Naan Mudhalvan – Generative AI
Project type	Concept design and prototype plan

A student-friendly project exploring how Generative AI can explain budgets and provide general spending insights.

1. Abstract

PocketSmart AI is a proposed personal budgeting assistant that helps users understand their income, expenses, and savings in a simple way. A user can enter a monthly income and record spending under categories such as food, travel, education, bills, and other needs. The system calculates totals and the remaining balance. A Generative AI component can then turn these figures into clear, general explanations and budgeting suggestions.

The project combines basic arithmetic and data organization with natural-language generation. It is intended for learning and planning support; it does not connect to bank accounts, make transactions, or provide professional financial advice. Users should verify their entries and decide for themselves which suggestions suit their circumstances.

2. Introduction

Managing money can be challenging when expenses are recorded in different places or are not reviewed regularly. A simple digital assistant can organize spending information and make it easier to see where money is going. PocketSmart AI presents this information through a clear summary and a conversational explanation.

Generative AI refers to systems that create new content, such as text, in response to a prompt. In this project, AI is used to explain calculated results in plain language. The calculations themselves should be performed by program logic, rather than relying on AI to do arithmetic.

3. Problem Statement

Many learners and households may find it difficult to track everyday expenses, compare spending with a budget, and understand how much money remains. Manual notes can be misplaced, and totals may be hard to review. The project addresses this organizational problem by providing one place to record expenses, calculate balances, and receive understandable summaries.

4. Aim

To design a beginner-friendly AI-assisted budgeting application that organizes income and expenses, calculates a remaining balance, and generates clear, general recommendations from the entered data.

5. Objectives

- Record a user-entered monthly income and expense entries.
- Group expenses into understandable categories.
- Calculate total expenses and remaining balance accurately using program logic.
- Show a simple summary that is easy to read.
- Use Generative AI to explain the summary in plain language and suggest general budgeting habits.
- Encourage privacy, careful review, and responsible use of AI.

6. Proposed System

The proposed system has four main parts:

Module	Purpose
Input module	Accepts monthly income, expense amount, date (optional), and category from the user.
Calculation module	Adds expense amounts and calculates balance: income minus total expenses.
AI explanation module	Receives a limited, non-identifying summary and creates a readable explanation or general suggestions.
Display module	Shows totals, category-wise spending, remaining balance, and AI-generated text.

7. Workflow

1. Enter data Income and expenses	2. Calculate Totals and balance	3. Summarize Prepare safe, limited data	4. Explain AI creates plain-language insights
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The application should validate inputs (for example, reject negative expense amounts) before calculating the summary. AI output should be treated as optional explanatory text and reviewed by the user.

8. Methodology

The project can be developed as a simple web or desktop prototype. The interface collects user-entered figures, while a calculation function produces exact totals. The AI feature can use a Generative AI model through an approved course environment, or be demonstrated with sample AI-generated text if a live model connection is not available.

Suggested development steps

- Plan the screen: income field, expense form, category selector, and summary panel.
- Validate entries and store only the information needed for the demonstration.
- Calculate total expenses and balance using a standard programming function.
- Create a short prompt containing only category totals and the balance; do not include names, bank details, passwords, or account numbers.
- Ask the AI to explain the summary in neutral, simple language, avoid guarantees, and offer general habits rather than personalized investment instructions.
- Test normal entries, empty fields, invalid numbers, and situations where expenses exceed income.

9. Sample Demonstration

The following values are fictional and are used only to demonstrate how the prototype may work.

Item	Amount (■)
Monthly income	10,000
Food	2,000
Travel	1,000
Education	2,000
Other expenses	2,000
Total expenses	7,000
Remaining balance	3,000

Example AI explanation

"Your entered income is ■10,000 and your recorded expenses total ■7,000, leaving a calculated balance of ■3,000. You may review the category totals and decide whether any planned costs can be adjusted. Keep some allowance for necessary and unexpected expenses. This is a general explanation based only on the sample figures."

10. Basic Algorithm

1. Read monthly income.
2. Read each expense amount and category.
3. Validate that the values are numeric and non-negative.
4. Add all valid expenses to obtain total expenses.
5. Calculate balance = income – total expenses.
6. Display the totals and category breakdown.
7. Send only the summarized figures to the AI feature, if enabled.
8. Display the generated explanation with a reminder to review it.

11. Expected Output and Benefits

The prototype is expected to present a compact budget summary with income, category-wise expenses, total expenses, and remaining balance. If an AI connection is available, it can add a short explanation in everyday language. This may help users review their spending records more consistently.

- Organization: keeps the entered budget details in one structured view.
- Clarity: converts totals into an easy-to-understand summary.
- Awareness: helps users notice which categories account for recorded expenses.
- Learning: demonstrates the combined use of programming, data handling, and Generative AI.
- Accessibility: can be designed with simple labels and readable screens.

12. Limitations

- The output is only as accurate as the information entered by the user.
- AI-generated text may be incomplete or misleading and must be checked.
- A prototype may not include secure long-term storage, multiple users, or bank integration.
- The project cannot know future expenses or a user's full financial circumstances.
- The system is an educational budgeting aid, not a financial adviser, accountant, or investment service.

13. Privacy and Responsible AI

The prototype should collect the minimum data required for its purpose. It should not request bank login details, card numbers, passwords, or other sensitive information. Where an AI service is used, the developer should follow the course platform's data rules and avoid sending personally identifying information. The interface should explain that AI-generated suggestions are general, may contain errors, and are not compulsory. Users remain responsible for reviewing their records and making their own choices.

14. Future Enhancements

- Add charts showing category-wise expenses over time.
- Allow users to set optional spending limits for categories.
- Offer downloadable monthly summaries.
- Support multiple languages and accessibility-friendly design.
- Add an offline demonstration mode with sample data.

15. Conclusion

PocketSmart AI demonstrates how Generative AI can be combined with simple calculations to make budget information easier to understand. By organizing income and expenses and explaining calculated results in plain language, the proposed assistant can support basic financial awareness. Careful input validation, privacy protection, and human review are essential to using the system responsibly.

References / Learning Resources

- Naan Mudhalvan – Generative AI course resource and project workspace (as provided by the learner).
- Google Cloud, "Generative AI" learning materials: <https://cloud.google.com/learn/what-is-generative-ai>
- Consumer Financial Protection Bureau, budgeting and money-management educational resources: <https://www.consumerfinance.gov/consumer-tools/>

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